Week 4
Series convergence test

F18CD Chap 3

Partial sums

Suppose (a_r), $r = 1, 2, \ldots$, is a sequence. Then we call

(a) $\sum_{k=1}^{\infty} a_k$ (infinite sum) a series;

(b) $\sum_{k=1}^{n} a_k = S_n$ (finite sum) a partial sums of the series in (a).

In addition, (S_n), $n = 1, 2, \ldots$, forms a sequence. In other words,

$$\left(S_n\right) = \left(a_1,\ a_1 + a_2,\ a_1 + a_2 + a_3, \ldots\right).$$

- Suppose $\left(S_n\right)$ converges to S, i.e., $\lim_{n\to\infty} S_n = S$, then the series $\sum_{k=1}^{\infty} a_k$ converges also. Furthermore, $\sum_{k=1}^{\infty} a_k = S$.
- Suppose the sequence $\left(S_n\right)$ diverges, then the series $\sum_{k=1}^{\infty} a_k$ diverges also.

This chapter focus on determining whether a give series is convergent or not.

Definition of Series

Let $S_n = \sum_{r=1}^{n} a_r$ where $\left(a_r\right)$ is a sequence. Then $\left(S_n\right)$ forms a sequence.

If $\left(S_n\right)$ converges, then $\sum_{r=1}^{\infty} a_r$ converges too. This is because

$$\lim\left(S_n\right) = \lim\left(\sum_{r=1}^{n} a_r\right) = \lim_{n\to\infty} \sum_{r=1}^{n} a_r = \sum_{r=1}^{\infty} a_r \, .$$

We call the infinite sum of a_r, i.e., $\sum_{r=1}^{\infty} a_r$ as a (convergent) series.

If $\left(S_n\right)$ diverges, then we say $\sum_{r=1}^{\infty} a_r$ a (divergent) series.

Definition of convergence series

Definition of convergence series:

The series $\sum_{r=1}^{\infty} a_r$ converges (to the limit L) if $\left(S_n\right)$ converges (to L).

Note: A series is convergence means the series is summable.

Note: Some authors call finite sum (as finite series) but in general series means infinite sum.

Example series that we can a formula for its finite sum (of first n terms)

We can determine the convergence of a series by taking the limit of $\sum_{r=1}^{n} a_r$ as $n \to \infty$.

$$\sum_{r=1}^{n} r = \frac{n}{2}(n+1) \Rightarrow \sum_{r=1}^{\infty} r = \lim_{n\to\infty}\left(\frac{n}{2}(n+1)\right) = \infty \text{ is a divergent series.}$$

$$\sum_{r=1}^{n} r^2 = \frac{n}{6}(n+1)(2n+1) \Rightarrow \sum_{r=1}^{\infty} r^2 = \lim_{n\to\infty}\left(\frac{n}{6}(n+1)(2n+1)\right) = \infty \text{ is a divergent series.}$$

$$\sum_{r=1}^{n}\frac{1}{r^2+r}=1-\frac{1}{n+1}\Rightarrow\sum_{n=1}^{\infty}\frac{1}{n^2+n}=\lim_{n\to\infty}\left(1-\frac{1}{n+1}\right)=1 \text{ is a convergent series.}$$

$$\sum_{r=1}^{n}2^r=2(2^n-1)\Rightarrow\sum_{r=1}^{\infty}2^r=\lim_{n\to\infty}\Big(2(2^n-1)\Big)=\infty \text{ is a divergent series.}$$

$$\sum_{r=1}^{n}\left(\tfrac{1}{2}\right)^r=1-\left(\tfrac{1}{2}\right)^n\Rightarrow\sum_{r=1}^{\infty}\left(\tfrac{1}{2}\right)^r=\lim_{n\to\infty}\left(1-\left(\tfrac{1}{2}\right)^n\right)=1 \text{ is a convergent series.}$$

This is the so-called Method 0 – Finding the explicit formula for the partial sum

Purpose of this chapter – What do we need to do?

We want to determine whether a given series is divergent or not.

But in general, it is hard to find the explicit formula (of partial sum).

Therefore, we need to explore – Series test of convergence.

Method 1: *n*-th term test (divergence test)

Theorem 4.2. If the series $\sum_{n=1}^{\infty} a_n$ converges, then $\lim\left(a_n\right) = 0$.

Contrapositive: If $\lim\left(a_n\right) \neq 0$ or $\lim\left(a_n\right) \nexists$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Remark: If $\lim\left(a_n\right) = 0$, there is no result. You cannot make any conclusion!!!

Example 4.3 (a)

Determine whether the series $\sum_{k=1}^{\infty} a_k$ is convergent or not when

(a) $a_k = (-1)^k$ (b) $a_k = \dfrac{k}{k+1}$ (c) $a_k = \dfrac{k}{k^2+1}$

Solution. We start with Method 1, check the limit of a_k when k tends to infinity (so you hope that the answer does not exist or not equal to 0).

(a) $\lim_{k\to\infty}(-1)^k \nexists$ so this implies that $\sum_{k=1}^{\infty} a_k$ is divergent.

Example 4.3 (b), (c)

(b) $a_k = \dfrac{k}{k+1}$ (c) $a_k = \dfrac{k}{k^2+1}$

(b) $\lim\limits_{k\to\infty} \dfrac{k}{k+1} = 1 \neq 0$ so this implies that $\sum\limits_{k=1}^{\infty} a_k$ is divergent.

(c) $\lim\limits_{k\to\infty} \dfrac{k}{k^2+1} = 0$ Further investigation needed.

Method 2: Geometric series test (Theorem 3.3)

Suppose we have a geometric series of the form $\sum_{r=0}^{\infty} x^r$ where x is a fixed real number. Then this series is

i. convergent if $|x| < 1$, and furthermore $\sum_{r=0}^{\infty} x^r = \frac{1}{1-x}$.

ii. divergent if $|x| \geq 1$.

Example 3.2

Example. Both $\sum_{r=0}^{\infty} 2^r$ and $\sum_{r=0}^{\infty} \frac{1}{2^r}$ are geometric series but we see that the former is divergent $(2 \geq 1)$ and the latter is convergent $\frac{1}{2} < 1$.

Check Example 3.2 for an independent proof of this example for the latter series.

Summary of Series test for convergence

When determine whether the series $\sum_{k=1}^{\infty} a_k$ is convergent:

Step 1: Is $\lim_{r\to\infty} a_r = 0$? Yes, then go to Step 2. No, then it is done.

Step 2: Is the series a geometric series? No, then go to Step 3. Yes, then it is done.

Method 3: *p*-series test (Theorem 4.6)

What is a p-series?

$\sum_{n=k}^{\infty} \frac{1}{n^p}$ is a p-series where k is any positive integer. Normally $k = 1$.

The p-series $\sum_{r=1}^{\infty} \frac{1}{r^p}$ converges if $p > 1$; and diverges if $p \leq 1$.

Example 3.4, Example 4.4

$\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\ldots+\frac{1}{n^2}+\ldots$ is a p-series with $p=2>1$, hence the series is convergent.

$1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\ldots+\frac{1}{n}+\ldots$ is a p-series with $p=1\leq 1$, hence the series is divergent (this series is called the Harmonic series).

$\sum_{n=1}^{\infty}\frac{1}{\sqrt{n}}=\sum_{n=1}^{\infty}\frac{1}{n^{\frac{1}{2}}}$ is a p-series with $p=\frac{1}{2}\leq 1$, hence the series is divergent.

Summary of Series test for convergence (updated)

When determine whether the series $\sum_{k=1}^{\infty} a_k$ is convergent:

Step 1: Is $\lim_{r\to\infty} a_r = 0$? Yes, then go to Step 2. No, then it is done.

Step 2: Is the series a geometric series? No, then go to Step 3. Yes, then it is done.

Step 3: Is the series a p-series? No, then go to Step 4. Yes, then it is done.

Geometry series vs p-series

$$\sum \text{constant}^{\text{variable}} \qquad \text{vs} \qquad \sum \frac{1}{\text{variable}^{\text{constant}}}$$

Recap

Let $\left(a_n\right)$ be a sequence and define $\left(s_n\right) := \left(a_1, a_1 + a_2, a_1 + a_2 + a_3\right)$.

Then $s_n = \sum_{k=1}^{n} a_k$ is the partial sum and $\sum_{k=1}^{\infty} a_k$ is called a series.

Note that $\boxed{\left(s_n\right) \text{ converges} \Leftrightarrow \sum_{k=1}^{\infty} a_k \text{ converges}}$

and furthermore $\boxed{\lim\left(s_n\right) = S = \sum_{k=1}^{\infty} a_k}$.

Some properties of convergent series: Theorem 3.5

Suppose $\sum_{r=1}^{\infty} a_r$ and $\sum_{r=1}^{\infty} b_r$ are convergent series such that $\sum_{r=1}^{\infty} a_r = A$ and $\sum_{r=1}^{\infty} b_r = B$. Then

1. $\sum_{r=1}^{\infty} \left(a_r + b_r\right)$ is convergent and $\sum_{r=1}^{\infty} \left(a_r + b_r\right) = A + B$.
2. $\sum_{r=1}^{\infty} \left(ca_r\right)$ is convergent and $\sum_{r=1}^{\infty} \left(ca_r\right) = cA$, where c is a fixed constant.

What about the multiplication/division of convergent series?

Let $\sum a_r$ and $\sum b_r$ be convergent. BUT

$\sum a_r b_r$ and $\sum \frac{a_r}{b_r}$ are not necessarily convergent!

Counter example: $\sum_{r=1}^{\infty} \frac{(-1)^{r+1}}{\sqrt{r}}$ and $\sum_{r=1}^{\infty} \frac{(-1)^{r}}{\sqrt{r}}$ by alternating series test

are convergence but $-\sum_{r=1}^{\infty} \frac{1}{r}$ and $-\sum_{r=1}^{\infty} 1$ are not.

What about the divergent series?

Worksheet 3, Q5.

Suppose series $\sum_{k=1}^{\infty} a_k$ diverges to ∞ and $r > 0$.

Then $\sum_{k=1}^{\infty} r a_k$ diverges to ∞.

Method 4: Comparison test (Theorem 4.7)

Let $0 \leq a_r \leq b_r \ \forall r \in \mathbb{N}$. Then $\sum_{r=1}^{\infty} b_r$ converges $\Rightarrow \sum_{r=1}^{\infty} a_r$ converges.

Contrapositive: $\sum_{r=1}^{\infty} a_r$ diverges $\Rightarrow \sum_{r=1}^{\infty} b_r$ diverges.

Remark: Normally we compare to the p-series.

Alternative Method 4: Limit comparison test

Let $a_r, b_r > 0 \ \forall r \in \mathbb{N}$ such that

$\left(\frac{a_r}{b_r}\right)$ converges to a limit $\neq 0$. Then

$$\sum_{r=1}^{\infty} a_r \text{ converges} \Leftrightarrow \sum_{r=1}^{\infty} b_r \text{ converges.}$$

Examples

Determine whether the series $\sum_{r=1}^{\infty} a_r$ is convergent or not when

(a) $a_r = \dfrac{r+4}{r^3+1}$ (b) $a_r = \dfrac{r^2}{r^3+1}$ (c) $a_r = \dfrac{(r+1)\sin^2 r}{r^3+3r+1}$

Solutions

(a) Note that $\forall r \in \mathbb{Z}^+$, $a_r = \dfrac{r+4}{r^3+1} \geq 0$ and

$$a_r = \frac{r+4}{r^3+1} < \frac{r+4r}{r^3} = \frac{5}{r^2} = b_r ,$$

where $\displaystyle\sum_{r=1}^{\infty} b_r = 5\sum_{r=1}^{\infty} \frac{1}{r^2}$ is a p-series with $p = 2 > 1$ (so it is convergent).

Hence, $\displaystyle\sum_{r=1}^{\infty} a_r$ is convergent (poorer than the poor).

Because if Σa_r converges then $\Sigma c a_r$ converges.

Alternative for (a)

(a) Alternative: $\lim_{r\to\infty} \frac{a_r}{b_r} = \lim_{r\to\infty} \left(\frac{r+4}{r^3+1} \div \frac{1}{r^2} \right) = \lim_{r\to\infty} \left(\frac{r^3+4r^2}{r^3+1} \right) = \lim_{r\to\infty} \left(\frac{1+\frac{4}{r}}{1+\frac{1}{r^3}} \right) = 1 \neq 0\,.$

Since $\sum_{r=1}^{\infty} b_r = \sum_{r=1}^{\infty} \frac{1}{r^2}$ is a p-series with $p = 2 > 1$ (so it is convergent),

$\sum_{r=1}^{\infty} a_r$ is convergent.

(b) First note that $\forall r \in \mathbb{Z}^+$, $a_r = \dfrac{r^2}{r^3+1} \geq 0$, and

$$a_r = \frac{r^2}{r^3+1} > \frac{r^2}{r^3+r^3} = \frac{1}{2r} = b_r,$$

where $\displaystyle\sum_{r=1}^{\infty} b_r = \frac{1}{2}\sum_{r=1}^{\infty}\frac{1}{r}$ is a harmonic series (it is divergent).

Hence, $\displaystyle\sum_{r=1}^{\infty} a_r$ is divergent (richer than the rich).

Alternative.

Choose $b_r = \frac{1}{r}$ then $\frac{a_r}{b_r} = \frac{r^2}{r^3+1} \div \frac{1}{r} = \frac{r^3}{r^3+1} \to 1 \neq 0$ when $r \to \infty$.

Hence $\sum_{r=1}^{\infty} a_r$ is divergent since $\sum_{r=1}^{\infty} b_r$ is divergent.

(c) First note that $\forall r \in \mathbb{Z}^+$, $a_r = \dfrac{(r+1)\sin^2 r}{r^3+3r+1} \geq 0$, and

$$a_r = \frac{(r+1)\sin^2 r}{r^3+3r+1} \leq \frac{r+1}{r^3+3r+1} \leq \frac{2r}{r^3} = \frac{2}{r^2} = b_r,$$

where $\displaystyle\sum_{r=1}^{\infty} b_r = 2\sum_{r=1}^{\infty} \frac{1}{r^2}$ is a p-series with $p = 2 > 1$ (so it is convergent).

Hence, $\displaystyle\sum_{r=1}^{\infty} a_r$ is convergent (poorer than the poor).

(c) Alternative method not very useful to be applied.

(c) Why alternative method is not helpful?

A reason to use limit comparison test is to "avoid" using Comparison test.

Now, $a_r = \dfrac{(r+1)\sin^2 r}{r^3+3r+1}$. What do we choose for b_r?

Choice 1: $b_r = \dfrac{1}{r^2}$. Then $\lim\left(\dfrac{a_r}{b_r}\right) = \lim \dfrac{\left(r^3+r^2\right)\sin^2 r}{r^3+3r+1} = \lim \dfrac{\left(1+\frac{1}{r}\right)\sin^2 r}{1+\frac{3}{r^2}+\frac{1}{r^3}} = ??$

Choice 2: $b_r = \dfrac{\sin^2 r}{r^2}$. Then $\lim\left(\dfrac{a_r}{b_r}\right) = \lim \dfrac{\left(r^3+r^2\right)}{r^3+3r+1} = \lim \dfrac{\left(1+\frac{1}{r}\right)}{1+\frac{3}{r^2}+\frac{1}{r^3}} = 1$. Ok, but is $\sum b_r$ a convergent or divergent series? We can show that it is convergent (by comparison test), then this defeats the purpose of "avoiding" the comparison test in the first place!!

When determine whether the series $\sum_{k=1}^{\infty} a_k$ is convergent:

Step 1: Is $\lim_{r\to\infty} a_r = 0$? Yes, then go to Step 2. No, then it is done.

Step 2: Is the series a geometric series? No, then go to Step 3. Yes, then it is done.

Step 3: Is the series a p-series? No, then go to Step 4. Yes, then it is done.

Step 4: Can we perform the Comparison test (compare with geometric series and p-series) to get the answer? No, then go to Step 5. Yes, then it is done.

Method 5: D'Alembert's Ratio test

Let $a_r > 0$ ($a_r \neq 0$) for all $r \geq 1$. Then check

$$L = \lim_{r \to \infty} \frac{a_{r+1}}{a_r}.$$

- If $L > 1$, then $\sum_{r=1}^{\infty} a_r$ is divergent.
- If $0 \leq L < 1$, then $\sum_{r=1}^{\infty} a_r$ is convergent.
- If $L = 1$, then the result is inconclusive.

The series diverges if $L = \infty$

Remark: Since $a_r > 0$ for all r, so it is impossible to have $L < 0$.

Example 4.17

Determine whether the series $\sum_{r=1}^{\infty} a_r$ is convergent or not when

(a) $a_r = \dfrac{r}{5^r}$ (b) $a_r = \dfrac{3^r}{r!}$ (c) $a_r = \dfrac{(r!)^2}{(2r)!}$

Solution.

(a) First we note that $\forall r \in \mathbb{Z}^+, \quad a_r = \dfrac{r}{5^r} > 0$. Then consider

$$L = \lim_{r\to\infty} \frac{a_{r+1}}{a_r} = \lim_{r\to\infty} \frac{\frac{r+1}{5^{r+1}}}{\frac{r}{5^r}} = \lim_{r\to\infty} \frac{r+1}{5r} = \frac{1}{5} < 1\,.$$

So we conclude that $\displaystyle\sum_{r=1}^{\infty} a_r$ is convergent.

(b) First we note that $\forall r \in \mathbb{Z}^+, \quad a_r = \dfrac{3^r}{r!} > 0$. Then consider

$$L = \lim_{r\to\infty} \frac{a_{r+1}}{a_r} = \lim_{r\to\infty} \frac{\frac{3^{r+1}}{(r+1)!}}{\frac{3^r}{r!}} = \lim_{r\to\infty} \frac{3}{r+1} = 0 < 1.$$

So we conclude that $\displaystyle\sum_{r=1}^{\infty} a_r$ is convergent.

(c) First we note that $\forall r \in \mathbb{Z}^+, \quad a_r = \dfrac{(r!)^2}{(2r)!} > 0$. Then

$$L = \lim_{r\to\infty} \frac{a_{r+1}}{a_r} = \lim_{r\to\infty} \frac{\frac{((r+1)!)^2}{(2r+2)!}}{\frac{(r!)^2}{(2r)!}} = \lim_{r\to\infty} \frac{(r+1)^2}{(2r+1)(2r+2)}$$

$$= \lim_{r\to\infty} \frac{(1+\frac{1}{r})^2}{(2+\frac{1}{r})(2+\frac{2}{r})} = \frac{1}{4} < 1\,.$$

So we conclude that $\sum_{r=1}^{\infty} a_r$ is convergent.

Self exercise

Determine whether the series $\sum_{r=1}^{\infty} a_r$ is convergent or not when

(a) $a_r = \dfrac{6^r}{r!}$ (b) $a_r = \dfrac{r3^r}{4^r}$ (c) $a_r = \dfrac{5^r}{4^r r}$ (d) $a_r = \dfrac{1}{r}$ (e) $a_r = \dfrac{\sqrt[3]{r}}{r!}$.

Solution.

(a) $a_r = \dfrac{6^r}{r!}$

$$\lim_{r\to\infty} \frac{\frac{6^{r+1}}{(r+1)!}}{\frac{6^r}{r!}} = \lim_{r\to\infty} \frac{6}{r+1} = 0 < 1 \Rightarrow \sum_{r=1}^{\infty} a_r \text{ converges.}$$

(b) $a_r = \dfrac{r3^r}{4^r}$

$$\lim_{r\to\infty} \frac{\frac{(r+1)3^{r+1}}{4^{r+1}}}{\frac{r3^r}{4^r}} = \lim_{r\to\infty} \frac{3(r+1)}{4r} = \frac{3}{4} < 1 \Rightarrow \sum_{r=1}^{\infty} a_r \text{ converges.}$$

(c) $a_r = \dfrac{5^r}{4^r r}$

$$\lim_{r\to\infty} \frac{\frac{5^{r+1}}{4^{r+1}(r+1)}}{\frac{5^r}{4^r r}} = \lim_{r\to\infty} \frac{5r}{4(r+1)} = \frac{5}{4} > 1 \Rightarrow \sum_{r=1}^{\infty} a_r \text{ diverges.}$$

(d) $a_r = \dfrac{1}{r}$

$$\lim_{r\to\infty} \frac{\frac{1}{r+1}}{\frac{1}{r}} = \lim_{r\to\infty} \frac{r}{r+1} = 1 \Rightarrow \text{the result is inconclusive.}$$

We note that this is a p-series ($p = 1$) and the series is divergent.

(e) $a_r = \dfrac{\sqrt{r}}{r!}$

$$\lim_{r\to\infty} \frac{\frac{\sqrt{r+1}}{(r+1)!}}{\frac{\sqrt{r}}{r!}} = \lim_{r\to\infty} \frac{\sqrt{1+\frac{1}{r}}}{r+1} = 0 < 1 \Rightarrow \sum_{r=1}^{\infty} a_r \text{ converges.}$$

Summary of Series test (updated again)

When determine whether the series $\sum_{k=1}^{\infty} a_k$ is convergent:

Step 1: Is $\lim_{r\to\infty} a_r = 0$? Yes, then go to Step 2. No, then it is done. (nth term test)

Step 2: Is the series a geometric series? No, then go to Step 3. Yes, then it is done.

Step 3: Is the series a p-series? No, then go to Step 4. Yes, then it is done.

Step 4: Can we perform the Comparison test (compare with geometric series and p-series) to get the answer? No, then go to Step 5. Yes, then it is done.

Step 5: Can we use Ratio test? (This is only applicable if a_r is a quotient.) Yes, then it is done.

Remark: Step 4 applicable when $a_r \geq 0$ for all r; whereas Step 5 applicable when $a_r > 0$ for all r.

Example 4.4. Harmonic series $\sum_{r=1}^{\infty} \frac{1}{r}$ is divergent.

Proof.

$$
\begin{aligned}
s_{2^n} &= 1 + \tfrac{1}{2} + \left(\tfrac{1}{3} + \tfrac{1}{4}\right) + \left(\tfrac{1}{5} + \tfrac{1}{6} + \tfrac{1}{7} + \tfrac{1}{8}\right) + \ldots + \left(\tfrac{1}{2^{n-1}+1} + \tfrac{1}{2^{n-1}+2} + \ldots + \tfrac{1}{2^n}\right) \\
&> 1 + \tfrac{1}{2} + \tfrac{1}{4} + \tfrac{1}{4} + \tfrac{1}{8} + \tfrac{1}{8} + \tfrac{1}{8} + \tfrac{1}{8} + \ldots + \underbrace{\left(\tfrac{1}{2^n} + \tfrac{1}{2^n} + \ldots + \tfrac{1}{2^n}\right)}_{2^{n-1} \text{ times}} \\
&= 1 + \underbrace{\tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{2} + \ldots + \tfrac{1}{2}}_{n \text{ times}} \\
&= 1 + \tfrac{n}{2} \to \infty \quad \text{as } n \to \infty
\end{aligned}
$$

Example 3.4. $\sum_{k=1}^{\infty} \frac{1}{k^2}$ is convergent.

Proof. Let $s_n = 1 + \frac{1}{4} + \frac{1}{9} + \ldots + \frac{1}{n^2}$.

Then $\left(s_n\right)$ is strictly increasing $\frac{1}{k^2} > 0 \ \forall k \in \mathbb{N}$.

Now, $s_n = 1 + \frac{1}{4} + \frac{1}{9} + \ldots + \frac{1}{n^2}$

$$< 1 + \frac{1}{2(1)} + \frac{1}{3(2)} + \ldots + \frac{1}{n(n-1)}$$

$$= 1 + \sum_{k=2}^{n} \frac{1}{k(k-1)} = 1 + \sum_{k=2}^{n} \left(\frac{1}{k-1} - \frac{1}{k}\right) = 2 - \frac{1}{n} < 2$$

Since $\left(s_n\right)$ is strictly increasing and bounded above,

$\left(s_n\right)$ is convergent by MCT. Hence, $\sum_{k=1}^{\infty} \frac{1}{k^2}$ is convergent.

Note that Method 4 and Method 5 deal only with non-negative and positive terms a_r. In general, a_r can be negative. In this section, we study the case a special type of convergent, called the *absolute convergence* of a series.

Definition. A series $\sum_{r=1}^{\infty} a_r$ is said to be absolutely convergent if $\sum_{r=1}^{\infty} |a_r|$ converges.

Method 6: Absolutely convergent

Theorem 4.12. $\sum_{r=1}^{\infty} |a_r|$ converges $\Rightarrow$ $\sum_{r=1}^{\infty} a_r$ converges.

Note that

1. $\sum_{r=1}^{\infty} a_r$ converges $\not\Rightarrow$ $\sum_{r=1}^{\infty} |a_r|$ converges.
2. $\sum_{r=1}^{\infty} |a_r|$ diverges $\not\Rightarrow$ $\sum_{r=1}^{\infty} a_r$ diverges.

Conditionally converge

Remark: For 1., there exists a series which is convergent but not it is not absolutely convergent. If this happens, we call the series converges conditionally.

Example: $\sum_{r=1}^{\infty} \frac{(-1)^{r+1}}{r}$ converges but $\sum_{r=1}^{\infty} \frac{1}{r}$ diverges. We say that $\sum_{r=1}^{\infty} \frac{(-1)^{r+1}}{r}$ converges conditionally.

We shall show that $\sum_{r=1}^{\infty} \frac{(-1)^{r+1}}{r}$ converges towards the end of the chapter.

Corollary

(a) (comparison test) If $\left|a_r\right| \le b_r \quad \forall r \ge 1$ and $\sum_{r=1}^{\infty} b_r$ converges. Then $\sum_{r=1}^{\infty} a_r$ converges.

(b) (ratio test) If $a_r \ne 0 \quad \forall r \ge 1$ and $\lim_{r\to\infty}\left|\frac{a_{r+1}}{a_r}\right| < 1$. Then $\sum_{r=1}^{\infty} a_r$ converges.

Summary of Series test

When determine whether the series $\sum_{k=1}^{\infty} a_k$ is convergent:

Step 1: Is $\lim_{r\to\infty} a_r = 0$? Yes, then go to Step 2. No, then it is done. (nth term test)

Step 2: Is the series a geometric series? No, then go to Step 3. Yes, then it is done.

Step 3: Is the series a p-series? No, then go to Step 4. Yes, then it is done.

Step 4: Can we perform the Comparison test (compare with geometric series and p-series) to get the answer? No, then go to Step 5. Yes, then it is done.

Step 5: Can we use Ratio test? (This is only applicable if a_r is a quotient.) Yes, then it is done. No, then go to Step 6.

Step 6 (only can be used to check that it is convergent): Is $\sum_{r=1}^{\infty} |a_r|$ converges? If yes, then $\sum_{r=1}^{\infty} a_r$ converges. If no, then it is not necessarily $\sum_{r=1}^{\infty} a_r$ diverges!

Example 4.14(a)

Consider the series $\sum_{r=1}^{\infty} \frac{(-1)^{r+1}}{r^2}$. Now $\sum_{r=1}^{\infty} \left| \frac{(-1)^{r+1}}{r^2} \right| = \sum_{r=1}^{\infty} \frac{1}{r^2}$ converges. So

$$\sum_{r=1}^{\infty} \frac{(-1)^{r+1}}{r^2} \text{ converges.}$$

Example 4.14(b)

Consider the series $\displaystyle\sum_{r=1}^{\infty} \frac{r \sin r}{r^3 + 5r^2 + 3}$. Now

$$\left| \frac{r \sin r}{r^3 + 5r^2 + 3} \right| \leq \frac{r}{r^3 + 5r^2 + 3} < \frac{r}{r^3} = \frac{1}{r^2},$$

and since $\displaystyle\sum_{r=1}^{\infty} \frac{1}{r^2}$ converges, we conclude that $\displaystyle\sum_{r=1}^{\infty} \frac{r \sin r}{r^3 + 5r^2 + 3}$ converges.

Example

Consider the series $\sum_{r=1}^{\infty} \frac{(-1)^r r}{3^r}$. Now

$$\lim_{r\to\infty} \left| \frac{(-1)^{r+1} \frac{r+1}{3^{r+1}}}{(-1)^r \frac{r}{3^r}} \right| = \lim_{r\to\infty} \frac{r+1}{3r} = \frac{1}{3} < 1.$$

By the ratio test, $\sum_{r=1}^{\infty} \left| \frac{(-1)^r r}{3^r} \right|$ converges, so we conclude that $\sum_{r=1}^{\infty} \frac{(-1)^r r}{3^r}$ converges.

Reveal the answers of Week 1 slide 6

- As we have seen in Q14 of Worksheet 2, permutation sequences preserves the convergence to the same limit.
- However, it is not always true in the case of series, the convergence is preserved if the series is absolutely convergence.
- Check Q16, Worksheet 4.

Theorem on Rearrangement (Riemann)

Let $f : \mathbb{N} \to \mathbb{N}$ be a permutation. Suppose $\left(a_n\right)$ and $\left(b_n\right)$ are sequences such that $b_{f(k)} = a_k$ for all $k \in \mathbb{N}$. Prove that if $\sum_{k=1}^{\infty} a_k$ converges absolutely, then $\sum_{k=1}^{\infty} b_k$ converges to the same limit.

The latter series is also known as the rearrangement of the former.

Method 7: Alternating series test

Suppose that

1. $a_r \geq 0$ for all $r \geq 1$;
2. (a_r) is a decreasing sequence, i.e., $a_r \geq a_{r+1}$ for all $r \geq 1$; and
3. $\lim_{r\to\infty} a_r = 0$.

Then the series $\sum_{r=1}^{\infty}(-1)^{r+1}a_r$ converges.

Remark: It can be shown that 2. and 3. will implies 1. So in general, we only need to check 2. and 3.

Remark: 1. Some authors write $\sum_{r=1}^{\infty}(-1)^r a_r$, $\sum_{r=0}^{\infty}(-1)^r a_r$, $\sum_{r=0}^{\infty}(-1)^{r+1} a_r$.

2. If one of the above hypotheses failed to comply, then the test is not applicable. Do not make conclusion that the alternating series is diverging.

Summary

When determine whether the series $\sum_{k=1}^{\infty} a_k$ is convergent:

Step 1: Is $\lim_{r\to\infty} a_r = 0$? Yes, then go to Step 2. No, then it is done.(nth term test)

Step 2: Is the series a geometric series? No, then go to Step 3. Yes, then it is done.

Step 3: Is the series a p-series? No, then go to Step 4. Yes, then it is done.

Step 4: Can we perform the Comparison test (compare with geometric series and p-series) to get the answer? No, then go to Step 5. Yes, then it is done.

Step 5: Can we use Ratio test? (This is only applicable if a_r is a quotient.) Yes, then it is done. No, then go to Step 6.

Step 6 (only can be used to check that it is convergent): Is $\sum_{r=1}^{\infty} |a_r|$ converges? If yes, then $\sum_{r=1}^{\infty} a_r$ converges. If no, then it is not necessarily $\sum_{r=1}^{\infty} a_r$ diverges!

Step 7 (for alternating series): Apply the alternating series test.

Example 4.10

Recall earlier we wish to show that $\sum_{r=1}^{\infty} \frac{(-1)^{r+1}}{r}$ converges. Now by applying the Alternating Series test; note that

1. $a_r = \frac{1}{r} \geq 0$ for all $r \geq 1$; (this condition is true if 2. and 3. is true.)
2. (a_r) is a decreasing sequence, i.e., $\frac{1}{r} \geq \frac{1}{r+1}$ for all $r \geq 1$; and
3. $\lim_{r\to\infty} a_r = \lim_{r\to\infty} \frac{1}{r} = 0$.

Then the series $\sum_{r=1}^{\infty} (-1)^{r+1} a_r = \sum_{r=1}^{\infty} \frac{(-1)^{r+1}}{r}$ converges.

More examples

Series	Series converges?	Converge absolutely?	Converge conditionally?
$\sum_{r=1}^{\infty} \frac{(-1)^r}{r}$	Yes	No	Yes
$\sum_{r=1}^{\infty} \frac{(-1)^r}{r^2}$	Yes	Yes	No
$\sum_{r=1}^{\infty} \frac{(-1)^r}{\sqrt{r}}$	Yes	No	Yes
$\sum_{r=1}^{\infty} \frac{(-1)^r}{\ln r}$	Yes	No	Yes
$\sum_{r=1}^{\infty} \frac{(-1)^r}{r!}$	Yes	Yes	No
$\sum_{r=1}^{\infty} (-1)^r$	No	No	No

There are only three possibilities for the following:

Series	Series converges?	Converge absolutely?	Converge conditionally?
Type 1	Yes $\Leftarrow$	Yes	No
Type 2	No $\Rightarrow$	No	No
Type 3	Yes	No	Yes

Extra series test: *n*-th root test

Let $a_r > 0 \; \forall r \in \mathbb{N}$. Then $\sum_{r=k}^{\infty} a_r$ converges if sequence $\left(\sqrt[r]{a_r}\right)$ converges to $L < 1$; the series diverges if $L > 1$ or $\left(\sqrt[r]{a_r}\right)$ diverges to positive infinity. The test is inconclusive if $L = 1$.

Example (a)

Example. Determine the convergence of the series:

(a) $\displaystyle\sum_{r=2}^{\infty} \frac{1}{(\ln r)^r}$ (b) $\displaystyle\sum_{r=1}^{\infty} \frac{2^{r+1}}{r^r}$ (c) $\displaystyle\sum_{r=1}^{\infty} \left(1+\frac{1}{r}\right)^r$ (d) $\displaystyle\sum_{r=1}^{\infty} \frac{5^r}{r^{r+1}}$

Solution.

(a) $\lim\left(\sqrt[r]{\frac{1}{(\ln r)^r}}\right) = \lim\left(\frac{1}{\ln r}\right) = 0 < 1$ so $\displaystyle\sum_{r=2}^{\infty} \frac{1}{(\ln r)^r}$ converges.

Example (b)

$$\text{(b)}\ \lim\left(\sqrt[r]{\frac{2^{r+1}}{r^r}}\right) = \lim\left(\frac{2\sqrt[r]{2}}{r}\right) = 0 < 1 \ \text{ so } \sum_{r=1}^{\infty} \frac{2^{r+1}}{r^r} \ \text{ converges.}$$

$$\text{Note that } \lim 2^{\frac{1}{r}} = 2^{\lim(\frac{1}{r})} = 2^0 = 1.$$

Example (c)

(c) $\lim\left(\sqrt[r]{\left(1+\frac{1}{r}\right)^r}\right) = \lim\left(1+\frac{1}{r}\right) = 1$ so the test is inconclusive.

However, $\lim\left(\left(1+\frac{1}{r}\right)^r\right) = e \neq 0$, from the divergence test, $\sum_{r=1}^{\infty}\left(1+\frac{1}{r}\right)^r$ diverges.

Note that the definition of $e^x := \lim\left(1+\frac{x}{r}\right)^r$ where x is a fixed real number.

Example (d)

(d) $\lim\left(\sqrt[r]{\frac{5^r}{r^{r+1}}}\right) = \lim\left(\frac{5}{r\sqrt[r]{r}}\right) = \lim\frac{5}{r}\lim\frac{1}{\sqrt[r]{r}} = 0 < 1$ so $\sum_{r=1}^{\infty}\frac{5^r}{r^{r+1}}$ converges.

Note that $\lim\left(\sqrt[r]{r}\right) = \lim r^{\frac{1}{r}} = \lim e^{\frac{\ln r}{r}} = e^{\lim\left(\frac{\ln r}{r}\right)} = e^0 = 1$.

Recall: $x = e^{\ln x}\ \forall x \in \mathbb{R}^+$.

Integral test

Suppose $k > 0$. Let f be a continuous, positive, decreasing function on $[1, \infty)$ and set $f(r) = a_r$. Then the series $\sum_{r=k}^{\infty} a_r$ converges iff the improper integral $\int_k^{\infty} f(x)\,dx$ is convergent.

Example

Show that the series $\sum_{r=1}^{\infty} \frac{\ln r}{r}$ is divergent.

Summary

Step 1. n-th term test: $\lim(a_r) \nexists$ or equal to nonzero, then $\sum_{r=1}^{\infty} a_r$ diverge. Otherwise, proceed to step 2.

Step 2. Geometry series: If $\sum_{r=1}^{\infty} a_r = \sum_{r=1}^{\infty} x^r$ for some fixed value $x \in \mathbb{R}$, then the series converges if $|x| < 1$ and diverges if otherwise.

Step 3. p-series: If $\sum_{r=1}^{\infty} a_r = \sum_{r=1}^{\infty} \frac{1}{r^p}$ for some fixed value $p \in \mathbb{R}$, then the series converges if $p > 1$ and diverges if otherwise.

Step 4. Comparison test. Let $a_r \leq b_r \, \forall r \in \mathbb{N}$. Then $\sum_{r=1}^{\infty} a_r$ converges if $\sum_{r=1}^{\infty} b_r$ converges.

Let $a_r \geq b_r \, \forall r \in \mathbb{N}$. Then $\sum_{r=1}^{\infty} a_r$ diverges if $\sum_{r=1}^{\infty} b_r$ diverges.

Step 5. We may apply limit comparison test also. Let $a_r, b_r > 0 \; \forall r \in \mathbb{N}$. If the sequence $\left(\frac{a_r}{b_r}\right)$ converges to limit not equal to zero, then $\sum_{r=1}^{\infty} a_r$ converges iff $\sum_{r=1}^{\infty} b_r$ converges.

Step 6. Ratio test. Let $a_r > 0 \; \forall r \in \mathbb{N}$. If the sequence $\left(\frac{a_{r+1}}{a_r}\right)$ converges to $L < 1$, then $\sum_{r=1}^{\infty} a_r$ converges. If $L > 1$ or $\left(\frac{a_{r+1}}{a_r}\right)$ diverges to positive infinity, then $\sum_{r=1}^{\infty} a_r$ diverges. The test is inconclusive if $L = 1$.

Step 7. n-th root test. Let $a_r > 0 \; \forall r \in \mathbb{N}$. Then $\sum_{r=k}^{\infty} a_r$ converges if sequence $\left(\sqrt[r]{a_r}\right)$ converges to $L < 1$; the series diverges if $L > 1$ or $\left(\sqrt[r]{a_r}\right)$ diverges to positive infinity. The test is inconclusive if $L = 1$.

Step 8. Integral test. Suppose $k > 0$. Let f be a continuous, positive, decreasing function on $[1, \infty)$ and set $f(r) = a_r$. Then the series $\sum_{r=k}^{\infty} a_r$ converges iff $\int_k^{\infty} f(x)\,dx$ converges.

For alternating series or series that has negative terms

Step 9. Absolute convergence test. $\sum_{r=1}^{\infty}\left|a_r\right|$ converges $\Rightarrow \sum_{r=1}^{\infty} a_r$ converges.

Note that if the series $\sum_{r=1}^{\infty} a_r$ converges but $\sum_{r=1}^{\infty}\left|a_r\right|$ diverges, then we say the series converges conditionally.

Step 10. Alternating series test. Let $\left(a_r\right)$ be decreasing and converges to 0, then $\sum_{r=1}^{\infty}(-1)^r a_r$ converges.

We ask, if the series is convergent, can we find the exact answer for the series?

Yes, for some cases. Here we list down some known exact answer for some convergent series!

1. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \ln 2$

2. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n-1} = \frac{\pi}{4}$

3. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{3n-2} = \frac{\pi\sqrt{3}}{9} + \frac{\ln 2}{3}$

4. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{4n-3} = \frac{\pi\sqrt{2}}{8} + \frac{\sqrt{2}\ln(1+\sqrt{2})}{4}$

5. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{3n-1} = \frac{\pi\sqrt{3}}{9} - \frac{\ln 2}{3}$

6. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$

7. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$

8. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^6} = \frac{\pi^6}{945}$

9. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$

10. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^4} = \frac{7\pi^4}{720}$

11. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^6} = \frac{31\pi^6}{30240}$

12. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$

13. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(2n-1)^4} = \frac{\pi^4}{96}$

14. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(2n-1)^6} = \frac{\pi^6}{960}$

15. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)^3} = \frac{\pi^3}{32}$

16. $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{\left\lfloor \frac{n-1}{2} \right\rfloor}}{(2n-1)^3} = \frac{3\pi^3\sqrt{2}}{128}$

17. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)} = \frac{1}{2}$

18. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+2)} = \frac{3}{4}$

19. $\displaystyle\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2(2n+1)^2} = \frac{\pi^2-8}{16}$

Prove that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

An ingenious proof!

Proof. Using the Taylor series,

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots \text{ for all } x \in \mathbb{R}$$

Taylor series will be covered in Chap 5

and note that $\sin x = 0$ when $x = n\pi$ for all $n \in \mathbb{N}$.

We write $\sin x$ in terms of its roots, i.e.,

$$\sin x = x\left(1 - \frac{x}{\pi}\right)\left(1 + \frac{x}{\pi}\right)\left(1 - \frac{x}{2\pi}\right)\left(1 + \frac{x}{2\pi}\right)\ldots = x\left(1 - \frac{x^2}{\pi^2}\right)\left(1 - \frac{x^2}{(2\pi)^2}\right)\ldots.$$

Compare these two equations and get the coefficient of x^3:

$$-\frac{1}{\pi^2} - \frac{1}{(2\pi)^2} - \frac{1}{(3\pi)^2} - \ldots = -\frac{1}{3!}$$

$$-\frac{1}{\pi^2}\left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots\right) = -\frac{1}{6}$$

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \ldots = \frac{\pi^2}{6}$$